AI-Enabled Fire/Smoke Detection and Automated Emergency Response System

Using Deep Learning and IoT · Phase-2 Project Review

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Introduction

Background and motivation

- Fire is a leading cause of property loss and casualties; the gap between ignition and human detection drives escalation.
- Point-type smoke/heat sensors are slow in open or high-ceiling spaces and cannot localise an event.
- CCTV is already installed almost everywhere, but humans cannot watch feeds reliably over long shifts.
- Deep learning (YOLO family) now runs in real time on commodity hardware — the camera itself can become the detector.
- Detection alone is not safety: alerts need confirmation, evidence, logging and a monitoring interface.

seconds

typical delay that decides escalation

0 → 1

cameras reused: no new sensors needed for vision

Objectives and scope

1 Real-time detector

YOLOv11n fire/smoke detector that runs on a laptop CPU, usable with existing CCTV/USB cameras.

2 Robust dataset

Stratified indoor + outdoor corpus with explicit negatives; augmentation for scale, lighting and scene variation.

3 Temporal confirmation

Convert noisy per-frame boxes into NORMAL → WARNING → CONFIRMED → RECOVERY states; never trigger on one frame.

4 Alerts and evidence

Snapshots, event database, buzzer, multi-channel notifications, acknowledgement and emergency stop.

5 Backend and dashboard

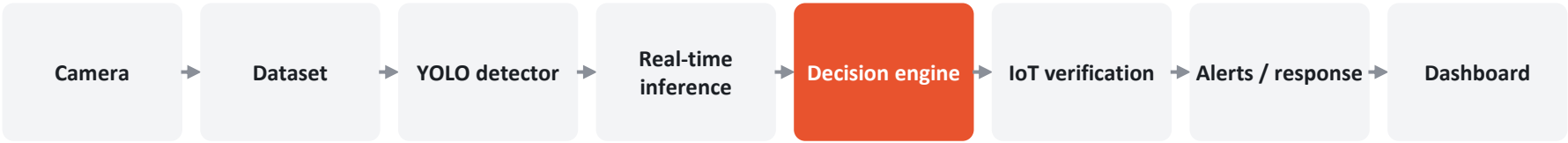
REST API + SQLite + live web dashboard for monitoring, history and device health.

6 Evaluation

Precision, recall, F1, mAP, false-alarm rate on negatives, CPU latency; fail-safe integration tests.

Scope: academic prototype with isolated low-voltage outputs — not certified fire-safety equipment.

Proposed solution and expected outcome



What we build

Vision-first detection on existing cameras · temporal confirmation with hysteresis · optional ESP32 gas/temperature fusion · evidence + events + notifications · live dashboard with acknowledge and emergency stop.

Expected outcome

Alert within seconds of a sustained flame/smoke, far fewer false alarms than single-frame or colour-rule detectors, a complete audit trail (who acknowledged, when, with evidence), and a reproducible model build with documented metrics.

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Problem Statement

Problem definition and existing gaps

Problem

Build a system that is simultaneously accurate on real CCTV-like footage, fast enough for live streams on commodity hardware, and embedded in a complete confirmation–alerting–monitoring pipeline.

1 Benchmark ≠ field

Published detectors are evaluated on curated images, not noisy, variably lit multi-angle CCTV — accuracy does not transfer.

2 Accuracy vs latency

Deeper nets and Transformer blocks gain accuracy at a latency cost that is unacceptable when seconds matter.

3 Detection ≠ safety

A raw bounding box cannot be operationalised without temporal confirmation, alerting, evidence and logging.

Our approach

Lightweight YOLOv11n fine-tuned on a stratified corpus with hard negatives → time-windowed state machine with separate enter/clear thresholds → IoT fusion → evidence, API, dashboard, fail-safes.

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Literature Survey

Review of relevant research

Method	Core technique	Year	Focus
YOLOv8/v7 fine-tuned (Saleh et al.)	single-stage detector	2024	wildfire smoke, flame
Improved YOLOv8s (Kong et al.)	light backbone + attention	2024	coal-mine fire/smoke
Improved YOLOv5s (Dou et al.)	feature-fusion refinement	2024	general fire
FS-YOLO (Wang et al.)	joint fire–smoke head	2024	fire and smoke
YOLOv11 (Khanam & Hussain)	C3k2, SPPF, C2PSA	2024	general detection
Shallow guide deep net (Li et al.)	lightweight CNN	2024	early-stage fire
Fire-ViT (Liu & Mariano)	Vision Transformer	2025	tunnel fire
CN2VF-Net / FireNet-KD (Ahmad et al.)	hybrid CNN–ViT, distillation	2025	complex scenes, wildfire
SmokeSeger (Jing et al.)	Transformer–CNN segmentation	2023	urban smoke
CAS-ViT (Zhang et al.)	additive self-attention ViT	2024	mobile vision

Comparison of methods and research gap

Two-stage (Faster R-CNN)

Highest localisation accuracy, but 5–17 fps: too slow for multi-camera live monitoring.

SSD / classic CNN classifiers

Fast, but frame-level only; weak on thin smoke and small flames.

YOLO family (v5 → v8 → v11)

Best accuracy-per-millisecond; anchor-free head and C2PSA attention help diffuse smoke. Chosen backbone.

ViT / hybrid (Fire-ViT, CN2VF-Net)

Strong in-domain accuracy; heavier inference, reduced robustness outside training domain.

Research gap identified

Reviewed work stops at per-frame metrics.

Rarely specified: how a detection becomes a verified alert; how steam, fog and lighting false alarms are suppressed over time; what happens when a camera, sensor or network fails.

Our contribution: robust false-alarm reduction + multi-stage automated response — not a new architecture.

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System Requirements

Hardware and software requirements

Hardware	Specification / role
Development laptop	Intel i5-1340P (12 cores), 16 GB RAM, no GPU — training and inference
Camera	USB webcam / IP / RTSP CCTV, 720p–1080p
ESP32	Sensor gateway + low-voltage actuator controller
MQ-2 gas, temperature sensor	Environmental evidence for fusion
Buzzer, relay, LED	Local alarm, isolated low-voltage prototype load

Software / tool	Use
Python 3.10+	all modules
PyTorch (CPU) + Ultralytics 8.4	YOLOv11n training/inference
OpenCV	capture, preprocessing, overlay
FastAPI + Uvicorn, SQLite	backend API, event database
HTML/CSS/JavaScript	dashboard (no framework)
Requests, SMTP, Telegram Bot API	notifications
pytest	23 unit + API tests
Git/GitHub, VS Code	version control, development

Dataset, API and other dependencies

Split	Images	Fire	Smoke	Both	Negative
Train	3,000	988	988	723	301
Validation	600	189	187	164	60
Test (held-out)	600	203	202	135	60

- Sources: Hugging Face fog_fire_detection (40.9k, CC-BY-4.0, outdoor + CCTV) and ISFire-4 (7.4k, MIT, indoor); 4 fine classes mapped to fire / smoke.
- Stratified per source and split; no video leakage between train and test; dataset register documents licence and mapping.

Backend API (FastAPI)

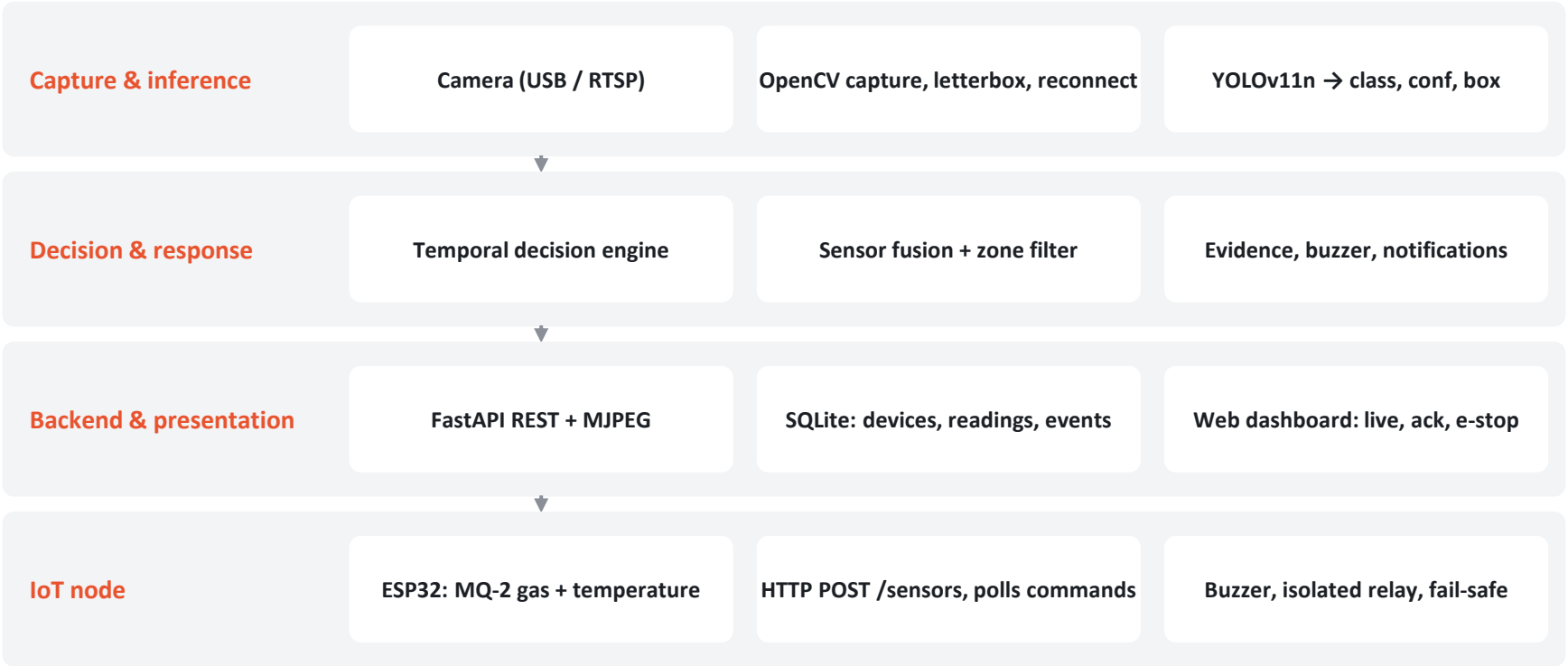
GET /health · GET /status
GET /devices · /devices/{id}
POST /sensors (ESP32)
POST /detections (AI)
GET /events · /events/{id}
POST /events/{id}/ack
POST /control/estop
GET /stream/{id} (MJPEG)

Config in YAML: thresholds, device, model, notifications; secrets in .env.

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System Design

System architecture and workflow



Modules and their functionalities

1 Dataset builder

training/prepare_dataset.py — stratified sampling from two zips, class remapping, data.yaml, manifest + register.

2 Trainer & evaluator

training/train.py, evaluate.py — YOLOv11n fine-tuning; per-class P/R/F1/mAP, FP/FN images, latency.

3 Camera & inference

vision/, inference/run_live.py — webcam/video/RTSP/folder, reconnect, overlay, evidence snapshots, keys.

4 Decision engine

decision_engine/ — time-window state machine, hysteresis, fusion, zone, ack, e-stop; 15 unit tests.

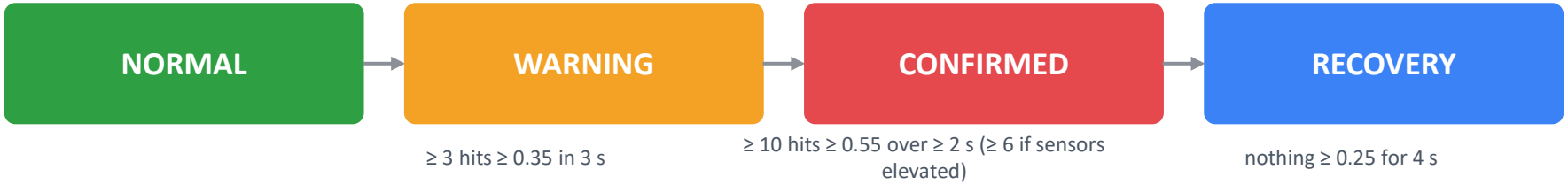
5 Backend & database

backend/ — FastAPI + SQLite; devices, sensors, detections, events, notifications; command queue; 8 API tests.

6 Dashboard, alerts, IoT

dashboard/, notifications/, iot/ — live UI, Telegram/e-mail, ESP32 firmware + sensor simulator.

Algorithm: temporal decision engine



- Sliding 3 s window of best-detection confidences; a single frame can never trigger.
- Separate enter (0.35 / 0.55) and clear (0.25) thresholds = hysteresis, no flapping.
- Sensor fusion: elevated gas/temperature lowers the confirmation bar; sensor-only evidence → SENSOR_WARNING.
- Stale sensor readings (> 15 s) ignored; acknowledge → RECOVERY; emergency stop blocks all physical outputs.
- RECOVERY → NORMAL after 10 s stable; re-appearance re-confirms immediately.

Also used

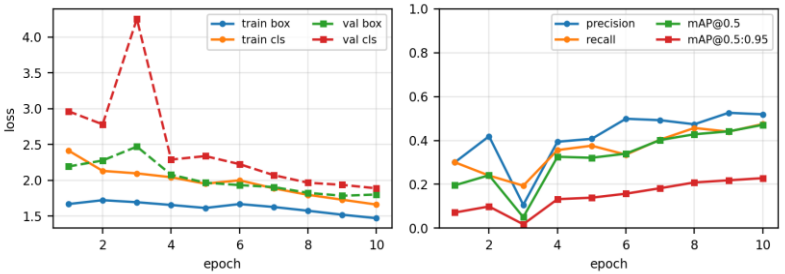
YOLOv11n (anchor-free, C2PSA) · IoU matching at 0.5 · NMS · mosaic/HSV/flip/scale augmentation · AdamW, seed 42 · precision/recall/F1/mAP@0.5/0.5:0.95.

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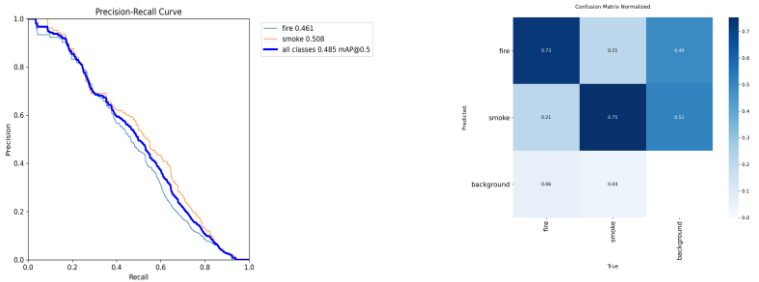
Implementation and Demonstration

Module 1 implemented: AI fire/smoke model

- Downloaded and merged two public datasets; stratified 3,000 / 600 / 600 split with 15 % negatives.
- YOLOv11n fine-tuned from COCO weights: 416 px, batch 16, 10 epochs, AdamW, seed 42 — on CPU (94.9 min).
- Evaluation script: per-class precision / recall / F1 / mAP, FP/FN example images, false-alarm rate on negatives, latency.
- Artifacts: models/best.pt, training_config.json, eval_report.md, PR curves, confusion matrix, dataset register.

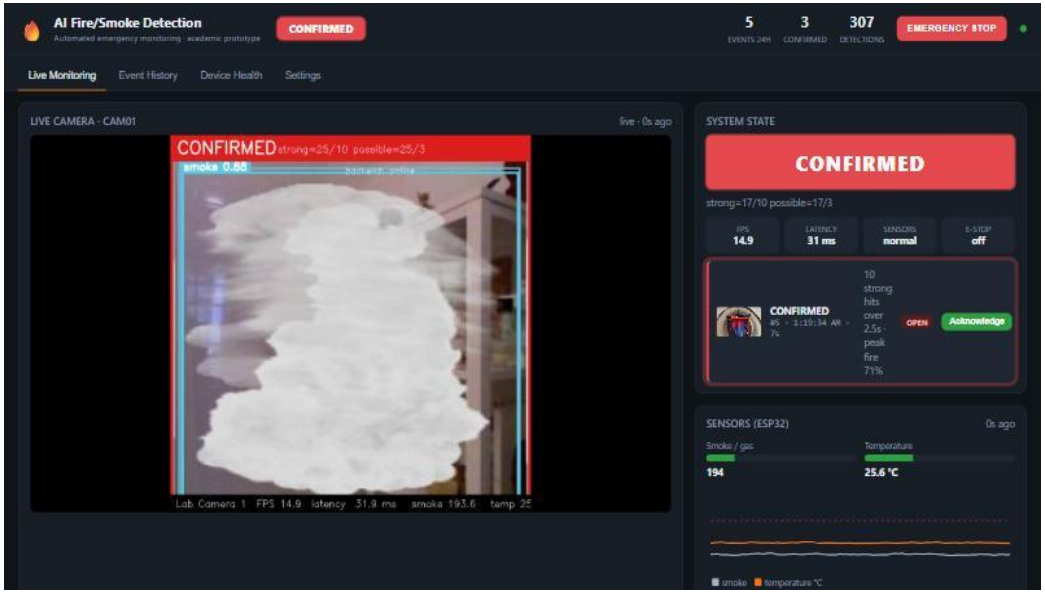


Training/validation loss and validation metrics per epoch



PR curves · normalised confusion matrix (test split)

Module 2 implemented: real-time pipeline



Live dashboard: annotated feed, state, sensors, latest detections, events

- Camera → YOLO → filters → time-window engine → overlay, evidence, buzzer, notifications.
- FastAPI + SQLite backend with the full required endpoint set; MJPEG live stream.
- Dashboard: NORMAL / WARNING / CONFIRMED, sensor gauges + chart, timeline, acknowledge, history, device health, e-stop.
- ESP32 simulator + firmware for sensor fusion; fail-safes for camera, sensor and network loss.
- 23 automated tests pass (engine + API); one-click setup/run scripts for any laptop.

Demonstration plan (two modules)

1 Model artefacts

Dataset register → training config → eval report, PR curve, confusion matrix, FP/FN examples.

2 Live NORMAL

Room only: green state, FPS/latency, sensors normal, device online.

3 Single-frame test

Flash a fire video < 1 s: boxes appear, state stays NORMAL.

4 WARNING → CONFIRMED

Hold fire: WARNING in ~1 s, CONFIRMED after 2 s — buzzer, evidence, event, notification.

5 Acknowledge / recovery

Dashboard Acknowledge → RECOVERY → NORMAL after 10 s; event closed with duration + peak.

6 Hard negative

Steam / white cloth / torch: no alarm.

7 Sensor fusion

Simulator smoke → SENSOR_WARNING; smoke+heat + fire → faster confirmation; disconnect → stale.

8 Fail-safes

Cover camera → CAMERA LOST + auto-reconnect; stop backend → inference continues; EMERGENCY STOP.

Results on the held-out test split

Class	Precision	Recall	F1	mAP@0.5	mAP@0.5:0.95
Fire	0.448	0.503	0.474	0.461	0.229
Smoke	0.590	0.476	0.527	0.508	0.265
All	0.519	0.490	0.504	0.484	0.247

- Operating point (conf 0.35, IoU 0.5): TP 346, FP 159, FN 681; the temporal engine filters isolated boxes before any alert.
- Engine verified by 15 unit tests; backend by 8 API tests; live fail-safe scenarios executed on the demo laptop.
- Numbers are a lower bound: 4,200-image subset, 10 epochs, no GPU — same scripts scale to the full 48k corpus.

0.484

mAP@0.5 (600 images, 10-epoch CPU model)

60/60

negative images with zero false alarms

31.4 fps

CPU only, 31.9 ms per frame at 416 px

Challenges encountered

No GPU

CPU-only training limited us to a 4,200-image subset and 10 epochs (~7 min/epoch); mitigated with YOLOv11n, 416 px, RAM caching, stratified sampling.

Dataset mismatch

Public sets are wildfire-heavy; indoor demo flames looked different. Merged an indoor set and balanced fire/smoke/both/negative.

False alarms

Reflections, torch light and steam produce single high-confidence boxes; solved by time-windowed confirmation and hysteresis, not by lowering thresholds.

Robustness

Camera disconnects, stale sensors and backend outages had to be handled explicitly (reconnect, staleness check, local buffering).

Shared CPU

Running live inference while training slowed both 10×; sequenced the work and added a headless/test mode.

Reproducibility

Config in YAML, fixed seed, dataset register, committed small weights so any laptop runs the demo without retraining.

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Paper Status

Paper title, target and current status

Title

AI-Based Fire and Smoke Detection System Using Deep Learning for Real-Time Emergency Monitoring

Target

IEEE-sponsored international conference (IEEE Xplore indexed); prepared in the IEEE conference (IEEEtran) two-column format, 7 pages.

Authors: M R Varun, M H Mayur Gowda, Kedar Lokesh Vandith; Dr. Janhavi V, Dr. Ayesha Taranum.

Current status

Full draft complete with measured results, literature table (12 works), architecture/workflow figures, algorithm and 28 references.

Next: guide review → final numbers from extended training → submission.

Thank you

Questions?

Repository: github.com/bhatspranav7/AI-Enabled-fire-Smoke-detection-systeem-using-Deep-learning-Emergency-Monitoring